

TOPICAL REVIEW

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Intraarterial Thrombolysis: Past, Present, Future

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ABSTRACT: Suboptimal clinical outcomes are common after ischemic stroke with large vessel occlusion despite current reperfusion treatment with intravenous thrombolysis and mechanical thrombectomy. Targeting residual distal arterial and microvascular occlusions with intraarterial thrombolysis has recently gained substantial interest as an adjunct to improve cerebral tissue perfusion. While recent trials provide promise in selected patients with anterior-circulation stroke, further high-quality, large-scale studies and pooled individual-patient analyses are needed. This review summarizes the evolving role of intraarterial thrombolysis over the past 3 decades and projects potential developments.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: humans ■ ischemic stroke ■ perfusion ■ reperfusion ■ thrombectomy



Timely restoration of cerebral blood flow is the primary aim of definitive treatment in ischemic stroke. Despite successful thrombectomy, a considerable proportion of previously well patients have poor clinical outcomes.¹ In the Highly Effective Reperfusion Evaluated in Multiple Endovascular Stroke meta-analysis of thrombectomy trials, 40% with extended Thrombolysis in Cerebral Ischemia (eTICI) 2c to 3 angiographic reperfusion did not retain functional independence at 90 days.² Residual distal arterial thrombi and persistent microvascular occlusion after macrovascular recanalization, the latter referred to as the no-reflow phenomenon, have gained increasing recognition as potential explanations and hence represent possible therapeutic targets to improve clinical outcomes in patients with acute large vessel occlusion (LVO).^{3,4} Multiple randomized clinical trials (RCTs) have sought to evaluate the efficacy and safety of adjunctive intraarterial thrombolysis (IAT) administered after successful thrombectomy in recent years.^{5–13} This review examines the historical origin, summarizes the contemporary research landscape, and projects future developments of IAT in ischemic stroke.

HISTORICAL ORIGIN

Primary IAT

Intraarterial access was the mode of administration when thrombolytics were first studied in stroke patients. In 1957, Sussman and Fitch first reported the use of intraarterial fibrinolysis with anterior-circulation stroke.¹⁴ Zeumer et al¹⁵ reported the use of intraarterial (IA) streptokinase for vertebrobasilar occlusion in Germany in 1983, which was followed by further reported cases with IA urokinase in middle cerebral artery occlusions.¹⁶

In 1998 and 1999, De Zoppo et al¹⁷ reported the first RCTs of IAT in ischemic stroke. The PROACT trial (Prolyse in Acute Cerebral Thromboembolism) and subsequent PROACT-II trial in 1999 found that more patients in the treatment arm had recanalization, functional independence, and intracerebral hemorrhage. A larger confirmatory trial was requested by the Food and Drug Administration for consideration of regulatory approval but was not undertaken due to cost. Nonetheless, primary IAT continued to be used off-label in patients with proximal middle cerebral artery occlusion. A systematic analysis in

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2002 including 27 studies and 852 patients who received IAT found more favorable outcomes, lower mortality rates, but more symptomatic intracranial hemorrhage (sICH) in patients receiving IAT.¹⁸ In the same year, the MELT trial (Middle Cerebral Artery Embolism Local Fibrinolytic Intervention) commenced in Japan but was later terminated prematurely following local regulatory approval of intravenous alteplase.¹⁹ While the trial did not meet its primary end point, IAT was subsequently listed as a recommended treatment in the Japanese clinical guideline.²⁰

Adjunctive IAT to Intravenous Thrombolysis

Several trials studied the combined use of IA and intravenous thrombolysis following Food and Drug Administration regulatory approval of intravenous alteplase in 1996. The EMS (Emergency Management of Stroke; 1999), IMS III (Interventional Management of Stroke; 2006–2012) and SYNTHESIS (Intraarterial versus Systemic Thrombolysis for Acute Ischemic Stroke; 2013) trials found no differences in outcomes or sICH.^{21–23} In 2015, the MR CLEAN trial (Multicenter Randomized Clinical Trial of Endovascular Treatment for Acute Ischemic Stroke in the Netherlands), conducted in the Netherlands, reported clear benefit of endovascular mechanical thrombectomy (MT) using a newer generation of stent retriever in patients with anterior-circulation LVO.²⁴ The results were confirmed by concurrent studies in other countries.¹ Following these landmark results, MT became the standard of care for acute LVO.

The use of IAT evolved to an adjunctive rescue intervention after unsuccessful MT with incomplete reperfusion or as a concurrent therapy to facilitate recanalization.²⁵ The 2019 American Heart Association/American Stroke Association guidelines listed the use of salvage technical adjuncts, inclusive of IAT, as a reasonable measure to achieve modified Thrombolysis in Cerebral Ischemia (mTICI) 2b/3 reperfusion. In an online survey of neurologists and neurointerventionalists in the German Stroke Registry, 39% of responders reported use of IAT on an individualized basis and 3% reported routine use in clinical practice.²⁶ Despite significant technical variations among operators in thrombectomy devices used, thrombolytic agent, and dosage administered, IAT appeared to be efficacious. A meta-analysis of 16 observational studies from 2009 to 2020 including 1348 patients found that patients treated with MT with adjunct IAT had better outcome compared with patients treated with MT only.²⁷ A registry from 10 European centers including 5612 screened patients also found rescue IAT to be associated with angiographic improvement and functional independence.²⁸

Angiographic-Clinical Mismatch in MT

Before the widespread use of MT, posttreatment arterial patency was commonly evaluated on follow-up computed

tomography (CT) angiogram performed at 24 to 48 hours. Digital subtraction angiography (DSA) enables periprocedural visualization of the macrovasculature to directly assess treatment efficacy. This recanalization assessment facilitated exploration of pathophysiological processes that may persist despite resolution of the primary vessel occlusion.

In the Highly Effective Reperfusion Evaluated in Multiple Endovascular Stroke analysis, 71% of patients achieved mTICI 2b/3 angiographic reperfusion, but only 46% of patients had a favorable clinical outcome at 3 months.¹ Contemporary observational and clinical trial studies observed a similar discrepancy between procedural success on angiography and clinical outcome.²⁹ Several factors have been proposed as an explanation for this angiographic-clinical mismatch, such as baseline frailty, established infarct before reperfusion, and poststroke complications. Among these, residual distal arterial thrombi and persistent microvascular occlusion, referred to as the no-reflow phenomenon, have gained substantial interest as potential therapeutic targets. Residual distal vessel occlusion, as identified by focal vascular hyperdensities on immediate postthrombectomy in-angio flat-panel detector CT, has been frequently reported in patients initially considered to have complete angiographic reperfusion.³⁰

Among patients without visually detectable residual occlusion, no-reflow has been detected in up to 25% of patients with eTICI 2c to 3 angiographic reperfusion immediately postthrombectomy and at 24 hours on CT or magnetic resonance imaging perfusion imaging.^{4,31}

CURRENT LANDSCAPE

With the growing recognition of angiographic-clinical mismatch and the no-reflow phenomenon, multiple trials in recent years have tested the use of IAT postthrombectomy in patients who were previously considered to have procedural success (Table). The phase 2b CHOICE trial (Chemical Optimization of Cerebral Embolectomy) randomized 121 patients with eTICI 2b to 3 angiographic reperfusion to receive postthrombectomy IA alteplase infusion and reported a higher rate of modified Rankin Scale (mRS) score of 0 to 1 at 90 days without an increase in sICH or mortality.⁵ The study was terminated early during the COVID-19 pandemic, which translated into a smaller-than-anticipated sample size with reduced precision of the effect estimate. Further limitations included a 30- to 40-minute delay in imaging, IVT, arterial puncture, and study drug initiation in the control group.

In the following 4 years, 7 prospective RCTs testing adjunctive IAT postthrombectomy from China were published. Five studies reported a neutral primary efficacy analysis. The ATTENTION-IA trial (Intraarterial Tenecteplase After Successful Endovascular Recanalisation in Patients With Acute Posterior Circulation

Table. xxx

Study	Design	Demo-graphics	IAT agent	Occlu-sion site	Angio-graphic criteria	Core size	Intra-venous throm-bolysis	Early anti-platelets	Primary efficacy outcome (IAT vs control)	Safety outcome (IAT vs control)
CHOICE ⁵	Phase 2b, Double-blinded placebo-control	Spanish	Alteplase 0.225 mg/kg, max 20 mg	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2b50$	ASPECTS >6	Permitted	Not per-mitted	mRS score of 0 to 1 59.0% vs 40.4% (aRD, 18.4% [95% CI, 0.3%–36.4%]; $P=0.047$)	slCH 0% vs 3.8% (RD, –3.8% [95% CI, –13.2 to 2.5%])
ATTENTION-IA ⁶	Phase 3, PROBE	Chinese	Tenecteplase 0.0625 mg/kg, max 6.25 mg	Basilar/ V ₄ /P1	eTICI $\geq 2b50$	PcASPECTS ≥ 7	Not per-mitted	Permit- ted	mRS score of 0 to 1 36.4% vs 26.0% (aRR, 1.36 [95% CI, 0.92–2.02]; $P=0.12$)	Mortality at 90 d 27.9% vs 26.9%, aRR 1.13, 0.73–1.74; slCH 8.3% vs 3.1% aRR 3.09, 0.78–12.20
POST-TNK ⁸	Phase 3, PROBE	Chinese	Tenecteplase 0.0625 mg/kg, max 6.25 mg	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2c$	ASPECTS ≥ 7	Not per-mitted	Permit- ted	mRS score of 0 to 1 49.1% vs 44.1% (aRR, 1.15 [95% CI, 0.97–1.36]; $P=0.11$)	Mortality at 90 d 16.0% vs 19.3% (aRR, 0.75 [95% CI, 0.50–1.13]; $P=0.16$); slCH 6.3% vs 4.4% (aRR, 1.43 [95% CI, 0.68–2.99]; $P=0.35$)
POST-UK ⁷	Phase 3, PROBE	Chinese	Urokinase 100 000 IU	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2c$	ASPECTS ≥ 7	Not per-mitted	Permit- ted	mRS score of 0 to 1 45.1% vs 40.2% (aRR, 1.13 [95% CI, 0.94–1.36]; $P=0.19$)	Mortality at 90 d 18.4% vs 17.3%; aRR, 1.06 [95% CI, 0.71–1.59]; $P=0.77$; slCH 4.1% vs 4.1% (aRR, 1.05 [95% CI, 0.45–2.44]; $P=0.91$)
ANGEL-TNK ¹⁰	Phase 3, PROBE	Chinese	Tenecteplase 0.125 mg/kg, max 12.5 mg	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2b50$	ASPECTS ≥ 6 CTP core <70 mL	Not per-mitted	Not per-mitted	mRS score of 0 to 1 40.5% vs 26.4% (RelR, 1.44 [95% CI, 1.06–1.95]; $P=0.02$)	slCH 5.6% vs 6.2% (RelR, 0.95 [95% CI, 0.36–2.53]; $P=0.92$); Mortality at 90 d 21.4% vs 21.7% (RelR, 0.76 [95% CI, 0.40–1.43]; $P=0.78$); any intracranial hem-orrhage within 48 h 24.6% vs 27.9% (RR, 0.90 [95% CI, 0.59–1.37]; $P=0.63$)
DATE ¹²	Phase 2a, Dose escalation/ PROBE	Chinese	Tenecteplase 0.0313 mg/kg, 0.0625 mg/kg (0.125 mg/kg dosage cohort excluded from com-parison with control)	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2b50$	ASPECTS ≥ 6	Not per-mitted	Not per-mitted	mRS score of 0 to 1 37.0% (0.0313 mg/kg) vs 33.8% (control), (aRR, 0.85 [95% CI, 0.54–1.35]; $P=0.50$); vs 43.5% (0.0625 mg/kg; aRR, 1.15 [95% CI, 0.73–1.80]; $P=0.55$)	slCH 4.3% (0.0313 mg/kg) vs 6.5% (0.0625 mg/kg) vs 3.1% (control)
PEARL ⁹	Phase 3, PROBE	Chinese	Alteplase 0.225 mg/kg, max 20 mg	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2b50$	ASPECTS ≥ 6	Permitted	Permit- ted	mRS score of 0 to 1 44.8% vs 30.2% (aRR, 1.45 [95% CI, 1.08–1.96]; $P=0.01$)	slCH 4.3% vs 5.0% (aRR, 0.85 [95% CI, 0.43–1.69]; $P=0.67$); mortality at 90 d 17.1% vs 11.3% (aRR, 1.60 [95% CI, 0.88–2.89]; $P=0.12$)
CHOICE III ¹³	Phase 3, PROBE	Spanish	Alteplase 0.225 mg/kg, max 20 mg	Intracra-nial ICA/ M1/prox M2	eTICI $\geq 2b50$	ASPECTS ≥ 7	Permitted	Not per-mitted	mRS score of 0 to 1 57.5% vs 42.9% (aRD, 15.0% [95% CI, 5.7%–24.3%]; $P=0.002$)	slCH 1.4% vs 0.5% (ARD, 3.10 [95% CI, 0.32–30.0]; $P=0.33$); mortality at 90 d 12.1% vs 6.4% (ARD, 5.9 [95% CI, 0.5–11.3]; $P=0.04$)

(Continued)

Table. Continued

Study	Design	Demo-graphics	IAT agent	Occlu-sion site	Angio-graphic criteria	Core size	Intra-venous throm-bolysis	Early anti-platelets	Primary efficacy outcome (IAT vs control)	Safety outcome (IAT vs control)
IAT-TOP ¹¹	Phase 3, PROBE	Chinese	Alteplase 0.225 mg/kg, max 22.5 mg	Basilar	eTICI ≥2b50	PcASPECTS ≥6	Permitted	Permitted	mRS score of 0 to 2 41.9% vs 46.7% (aRR, 0.93 [95% CI, 0.73–1.18]; <i>P</i> =0.55)	Mortality at 90 d 29.6% vs 27.0% (aRR, 1.07 [95% CI, 0.71–1.61]; <i>P</i> =0.75); sICH 2.4% vs 2.5% (RR, 0.98 [95% CI, 0.20–4.74]; <i>P</i> =0.97)
TECNO ³²	Phase 3, PROBE	European	Tenecteplase 3 mg	Intracranial ICA/M1/M2/M3/A1/A2/P1/P2	FCA/M1: eTICI 2b–c; For A1/A2, P1/P2, M2/M3: eTICI 2a–c; emboli to new territory	CT-ASPECTS ≥5; DWI-ASPECTS ≥4	Permitted	Permitted	Early reperfusion 39% vs 36% (aRD, 3.4% [95% CI, –15% to 21.7%]; <i>P</i> =0.68); late reperfusion 51% vs 40% (ARD, 15.4% [95% CI, –3.3% to 32.9%]; <i>P</i> =0.069)	sICH 5.2% vs 8.6%, <i>P</i> =0.54

aRD indicates adjusted risk difference; aRR, adjusted risk ratio; ASPECTS, Alberta Stroke Program Early Computed Tomography Score; eTICI, extended Thrombolysis in Cerebral Ischemia; IAT, intraarterial thrombolytic; mRS, modified Rankin Scale; PROBE, prospective, open-label, blinded end point; ReIR, relative risk; and sICH, symptomatic intracerebral hemorrhage.

Arterial Occlusion) randomized 204 patients with eTICI 2b to 3 reperfusion after MT for posterior-circulation LVO to treatment with IA tenecteplase 0.0625 mg/kg or MT alone.⁶ POST-TNK (Adjunctive Intraarterial Tenecteplase Following Near-Complete to Complete Reperfusion for Large Vessel Occlusion Stroke) randomized 504 patients with anterior-circulation LVO with eTICI 2c to 3 reperfusion to 0.0625 mg/kg tenecteplase postthrombectomy IAT or MT alone.⁸ In parallel, POST-UK randomized 534 patients to test IAT urokinase 100 000 IU.⁷ The phase 2a DATE trial (Safety and Efficacy of Adjunctive Intraarterial Tenecteplase Following Successful Thrombectomy in Patients With Large Vessel Occlusion) randomly assigned 157 patients to adjunctive IAT tenecteplase at 0.0313 mg/kg, 0.0625 mg/kg, or MT alone in patients with eTICI 2b to 3 reperfusion after MT for anterior-circulation LVO.¹² Finally, the IAT-TOP trial (Intraarterial Alteplase Thrombolysis After Successful Thrombectomy for Acute Ischemic Stroke in the Posterior Circulation) tested IAT alteplase (0.225 mg/kg) in patients with acute basilar occlusion with TIBI 2b to 3 angiographic reperfusion.

Conversely, 2 trials found a greater likelihood of excellent outcomes at 90 days in patients receiving adjunct IAT. The ANGEL-TNK trial (Intraarterial Recombinant Human TNK-tPA Thrombolysis for Acute LVO After Successful Mechanical Thrombectomy Recanalization) recruited 255 patients to test adjunct IAT with 0.125 mg/kg tenecteplase in anterior-circulation LVO after eTICI 2b to 3 reperfusion.¹⁰ The PEARL trial (Intraarterial Alteplase for Acute Ischemic Stroke After Mechanical Thrombectomy) included 324 patients with anterior-circulation LVO

and eTICI 2b to 3 angiographic reperfusion who were assigned to adjunct IAT with 0.225 mg/kg alteplase or MT alone.⁹

Multiple study-level meta-analyses of recent IAT trials found supportive evidence for the efficacy and safety of postthrombectomy IAT for anterior-circulation stroke, but not posterior-circulation stroke. However, these meta-analyses noted significant heterogeneity in the design, patient population, and procedural metrics across studies, which may contribute to the variation in treatment effect reported across trials.^{33–36} The predominance of Asian patients also limited international generalizability. Two recent European trials testing IAT presented further mixed results. The CHOICE 2 trial reported improved rates of excellent functional outcome when comparing 214 patients with anterior-circulation LVO receiving adjunct IAT alteplase with 219 patients with MT alone.¹³ However, the phase 3 TECNO trial (Safety and Efficacy of Intraarterial Tenecteplase for Noncomplete Reperfusion of Intracranial Occlusions) tested the effect of IA bolus tenecteplase in patients with incomplete reperfusion after MT and found no differences in the coprimary perfusion imaging and secondary clinical end points.³²

Ongoing Uncertainties and Unaddressed Questions

To date, 10 RCTs of IAT have been completed in the last 5 years, of which 4 studies met their primary end points. Within these positive trials, the emerging signal is strongest for excellent outcomes among patients already likely to achieve a good recovery, but evidence that IAT

reduces death or disability remains limited. Furthermore, several critical questions remain unanswered regarding the mechanism, application, and patient selection for adjunctive IAT.

Mechanism of Action

A fundamental question relates to the mechanism of action of adjunctive IAT in patients with angiographic reperfusion. While improving tissue reperfusion is the common aim, only CHOICE, ANGEL-TNK, PEARL, and CHOICE 2 performed follow-up perfusion imaging. All 4 trials enrolled patients with eTICI 2b angiographic reperfusion and applied a delay-based perfusion metric to quantify persistent hypoperfusion. A secondary analysis of 36 patients in the CHOICE trial reported fewer visible $T_{\max} > 6$ -second lesions at 48 hours in the IAT group.³⁷ The CHOICE 2 trial, which was designed to validate the findings of the CHOICE trial, also reported similar findings, albeit using an alternative metric (presence of a delay time > 3 -second lesion of ≥ 10 contiguous voxels). However, the ANGEL-TNK and PEARL trials found no differences in tissue reperfusion, which was assessed by $T_{\max} > 6$ -second lesion volume at 24 hours between groups, despite clinical improvement in the treatment arm.

Distinguishing Between Distal Arterial Occlusion and No-Reflow

Persistent hypoperfusion, reported in 63% of patients with mTICI 2b and 42.5% of patients with mTICI 3 angiographic reperfusion, may be due to 2 distinct entities with different pathological bases and radiological characteristics: residual distal arterial thrombi or the no-reflow phenomenon (Figure S1).^{38–40} As no-reflow can only be evaluated in patients with macrovascular reperfusion that is operationally defined as eTICI 2c to 3, inclusion of patients with eTICI 2b complicates the assessment of the mechanism of action of adjunctive IAT. In CHOICE 2, 18% of patients who achieved final eTICI 3 reperfusion in the absence of $> 70\%$ macrovascular stenosis or vasospasm exhibited hypoperfusion on follow-up CTP (unpublished). Nevertheless, persistent hypoperfusion in patients without visible macrovascular occlusion on DSA may still be due to residual arterial occlusion, as up to 15% of patients with core laboratory-adjudicated eTICI 3 reperfusion have been shown to have occult distal arterial occlusions on flat-panel detector CT acquired immediately postthrombectomy.³⁰ Importantly, $T_{\max} > 6$ seconds and delay time > 3 seconds primarily characterize a prolonged transit of blood reaching the tissue bed through collaterals and were designed and validated for use only in the prerevascularization setting to reflect macrovascular arterial occlusions upstream.⁴¹ In contrast, the anatomic site of occlusion in no-reflow is considered to be

within the microvasculature in the setting of unimpeded delivery of blood through the proximal arterial vasculature. The presence of a wedge-shaped $T_{\max} > 6$ -second or delay time > 3 -second abnormality therefore does not align with the pathophysiological understanding of the no-reflow phenomenon. At present, the precise physiological meaning of radiological perfusion abnormalities postthrombectomy remains debated.^{39,42} An international Delphi survey that aims to provide a consensus imaging definition and framework to systematically report no-reflow postthrombectomy is in progress.

Distinguishing between residual distal arterial occlusion and no-reflow has therapeutic implications. While arterial and microvascular in situ thrombosis may be addressed by fibrinolytic agents, no-reflow involves additional pathophysiological processes, such as pericyte contraction, endothelial injury, inflammation implicating reactive oxygen species, vascular remodeling and neutrophil plugging, which may not respond to fibrin-targeted treatment.⁴³ These factors may partly explain why approximately half of the recruited patients in the treatment arm remain disabled at 90 days among recent IAT studies.³³



Procedural and Performance Bias

A more provocative question is whether thrombolysis is the only therapeutic benefit received by patients assigned to the treatment group in recent IAT trials. Except for the CHOICE trial, all IAT trials that reported clinical efficacy were open-label in design, in which treatment was administered as a 15- to 30-minute infusion. Patients assigned to the control arm did not receive any endovascular intervention. Animal models of middle cerebral artery occlusion in rats have shown that IA infusion of isotonic saline alone improved cerebral blood flow and reduced leukocyte infiltration, vascular plugging, and infarct volume.⁴⁴ Concordantly, clinical data from cardiology trials have found improvement in ST-segment resolution, coronary flow restoration, and clinical outcome following IA injection of saline in patients with documented coronary no-reflow after percutaneous coronary intervention.⁴⁵ The impact of potential procedural biases from additional microcatheter placement for drug administration, and further rescue retrieval attempts after IAT, has also not been well explored in recent IAT trials. To address these potential confounders, a blinded, placebo-controlled design should be considered in future trials.

Efficacy

Meta-analyses of recent IAT trials have found a higher proportion of patients with mRS score of 0 to 1 at 90 days in the treatment arm. However, improvement in secondary clinical outcomes (such as the proportion of patients with mRS score of 0 to 2, ordinal shift on mRS

score, and quality-of-life metrics) and radiological outcomes (such as perfusion deficit and infarct volume) have not been consistently demonstrated. CHOICE 2 was the only trial that reported improvement in the self-reported quality-of-life metric. Clinical improvement without a corresponding reduction in infarct volume is well reported and may relate to intralesional salvage within the infarct lesion, which may be insensitive to conventional imaging metrics.^{37,46,47} However, discordance between the primary efficacy clinical end point of mRS score of 0 to 1 and most secondary clinical end points, which has been observed repeatedly in the recent trials, remains unexplained. Several factors may explain this, including ceiling effects after successful thrombectomy and a larger sample size required to detect differences in poor outcomes. Severe disability is driven by multifactorial and often irreversible mechanisms. Consequently, poor outcomes may be less modifiable despite successful reperfusion.

Although an incremental functional improvement from an mRS score 2 category (mild disability) to an mRS score 0 to 1 category (no disability despite mild symptoms) can be clinically meaningful to patients who were already destined to achieve a relatively favorable outcome, this does not address the broader conundrum of persistent poor outcome (death or disability, ie, mRS score of 3–6) despite successful thrombectomy, which affects more patients and incurs a substantially higher societal economic burden.⁴⁸ Despite being the underlying rationale and primary objective across recent IAT trials, most have not been powered to assess differences in death or disability postthrombectomy (mRS score of 0–2 versus 3–6) in the primary analysis.

Safety

The incidence of sICH was a common primary safety end point across recent IAT trials. Assessment of sICH was performed using the Heidelberg criteria in the 5 Chinese trials, while DATE, CHOICE, and CHOICE 2 applied the ECASS-3 criteria. Trials varied in inclusion criteria relevant to bleeding risk. The POST-UK (Adjunctive Intraarterial Urokinase After Near-Complete to Complete Reperfusion for Acute Ischemic Stroke), POST-TNK, ATTENTION-IA, IAT-TOP, and TECNO trials permitted stenting and angioplasty with concomitant antiplatelet use. The CHOICE, PEARL, and CHOICE 2 trials included patients receiving IVT. Despite these variations, the rates of sICH were similar when comparing treatment groups in the recent trials. However, while ANGEL-TNK reported a nonsignificant reduction in sICH in patients receiving the highest dose of TNK across trials (0.125 mg/kg), the DATE trial reported a significantly greater rate of sICH (25%) using the same dose among patients with similar characteristics.

Finally, beyond hemorrhagic safety end points, a key concern is the reported higher rate of mortality in PEARL

(17.2% versus 11.3%; $P=0.17$), TECNO (33.3% versus 21.8%; 95% CI, -0.5% to 27.0%), and CHOICE 2 (12.1% versus 6.4%; $P=0.04$). A meta-analysis of 6 IAT trials conducted in 2025 reported identical rates of mortality between groups with no heterogeneity across studies, but did not include the latter 2 studies. There was a notable increase in early deaths in the first week in the IAT group ($n=10$ versus 2) in CHOICE 2. Half of these were attributed to malignant cerebral edema, which is uncommon among patients with isolated intracranial occlusion presenting within 6 hours with a small baseline ischemic core who achieved timely and successful reperfusion.⁴⁹ Whether these adverse effects relate to iatrogenic blood-brain barrier breakdown induced by alteplase warrants further and careful investigation.^{50,51} While there was no increase in hemorrhagic complications in recent IAT trials, postthrombectomy cerebral edema development independent of parenchymal hematoma and sICH has been reported in the Highly Effective Reperfusion Evaluated in Multiple Endovascular Stroke meta-analysis and large-core thrombectomy trials.^{52,53}

Optimal Dose and Technical Approach

The technical approach for thrombolytic administration was similar across the published trials and comprised an infusion through a microcatheter proximal to or at the site of the initial occlusion after angiographic reperfusion. The CHOICE 2 trial empirically infused alteplase at the ICA terminus. In TECNO, $>50\%$ of the injection was made within 3 cm of the proximal residual clot face to facilitate first-pass effect through a higher local thrombolytic concentration.

Thrombolytic agents studied differed between the trials. Three of the 4 trials that met primary efficacy end points tested alteplase at 25% of the corresponding systemic IVT dose, in keeping with the earlier IMS III (alteplase 22 mg maximum) and EMS (alteplase 20 mg maximum) trials.^{21,22} Tenecteplase has pharmacological advantages of higher fibrin specificity and ease of bolus administration. The DATE trial tested 3 dosages of tenecteplase infusion and reported safety only at the lower tiers of 0.03125 mg/kg (1/8 of systemic dose) and 0.0625 mg/kg (1/4 of systemic dose). The TECNO trial examined a fixed 3 mg bolus of tenecteplase, which approximates 0.0375 mg/kg for an 80 kg patient, and found no difference in the rate of reperfusion. In contrast, the ANGEL-TNK trial reported efficacy and safety at a 0.125 mg/kg dosage (1/2 of systemic dose). Dose optimization has not been performed using alteplase or urokinase.

Patient Selection

To date, adjunctive IAT after successful MT has not been found to improve outcomes in posterior-circulation LVO,

and the ideal patient population with anterior-circulation LVO remains to be determined. The neutral POST-TNK and POST-UK trials limited inclusion to patients with near-complete angiographic reperfusion (TICI 2c–3). In comparison, trials that reported efficacy included patients with TICI 2b reperfusion. A subgroup analysis of ANGEL-TNK suggested the benefit of IAT was limited to patients with TICI 2b reperfusion, in concordance with TICI subgroup analysis in meta-analyses.¹⁰ However, the CHOICE 2 study reported a greater effect size favoring IAT in the 2c/3 compared with the TICI 2b group instead. This mixed finding presents in the setting of likely variations in periprocedural TICI assessment between operators, recruiting sites, trials, and core-laboratories.

The impact of pre-MT IVT on the efficacy of IAT is also unclear. The presence of thrombolytics in the systemic circulation may limit the observed benefits of additional agents administered locally. The ATTENTION-IA, POST-UK, POST-TNK, and ANGEL-TNK trials excluded patients who received IVT. Among the 4 trials that permitted enrollment of patients with anterior-circulation stroke who received IVT, subgroup analyses have consistently demonstrated a smaller effect size of IAT in patients with IVT.

Furthermore, stroke cause may influence treatment effect. Exploratory subgroup analyses in the POST-UK, POST-TNK, and IAT-TOP trials reported greater effect size in cardioembolic strokes. A summary of IAT efficacy for different patient populations is provided in [Table S1](#).

Unaddressed Patient Groups and Indications

Recent trials of adjunctive IAT included a selected cohort of patients who represent a minority of patients undergoing MT in practice. Among 2061 patients receiving MT who were screened for the ANGEL-TNK trial, only 256 patients were included. Notable exclusions include 654 patients requiring antiplatelets or anticoagulation within 24 hours of treatment, 507 patients who received bridging IVT, 56 patients with unsuccessful thrombectomy, and 72 patients presenting with large ischemic core.¹⁰ In the PEARL trial, which permitted IVT pretreatment, 313 patients who required endovascular stenting were excluded among 1879 patients screened.⁹ In the CHOICE trial, 121 patients were randomized after excluding 275 patients with unsuccessful thrombectomy and 109 patients who required stenting with DAPT.

Patients requiring concomitant or early antiplatelet therapy for endovascular stenting were excluded in the 4 positive IAT trials. A few older studies provide some insights regarding the use of IAT among patients with tandem occlusion who require stenting and early administration of antiplatelet agents. Lavellee et al⁵⁴ reported a series of 10 patients with tandem occlusion in which all 6 patients who received carotid wall stent, IAT, preprocedural abciximab infusion and postprocedural dual

antiplatelet therapy (clopidogrel 300 mg and aspirin 250 mg immediately postprocedure) had recanalization and clinical improvement without sICH. Kwon et al⁵⁵ reported 16 cases of adjunctive IA tirofiban after unsuccessful primary IAT with urokinase, in which 13 patients achieved angiographic recanalization and 2 patients developed sICH. Mangiafico et al⁵⁶ reported a case series of 21 patients with LVO who received intravenous tirofiban followed by IAT urokinase. Immediate recanalization was reported in 17 patients; however, 3 developed fatal sICH.

Patients presenting with large ischemic core comprise up to 33% of thrombectomy patients in real-world practice⁵⁷ but were uniformly excluded from adjunctive IAT trials. The ANGEL-TNK and PEARL trials had the most inclusive enrollment criteria for baseline core extent and permitted recruitment with a minimum Alberta Stroke Program Early Computed Tomography Score (ASPECTS) of 6. Nonetheless, the median ASPECTS of recruited patients in both trials were high (median ASPECTS 9 [interquartile range, 7–10] in ANGEL-TNK and median ASPECTS 7 [interquartile range, 6–8] in PEARL). Notably, median baseline core volume of patients in the ANGEL-TNK trial was <20 mL. To date, there is a paucity of data on the safety and efficacy of post-MT IAT in patients presenting with large core.

Finally, the role of IAT as rescue therapy after unsuccessful thrombectomy, which occurs in every tenth patient, has yet to be comprehensively studied in prospective RCTs in the modern stent-retriever era. The TECNO trial aimed to assess the use of IAT in patients with noncomplete reperfusion, but patients with eTICI 0 to 2a comprised only ≈11% (n=17/159) of the overall study population. Early cohort studies and registries have suggested that rescue IA urokinase seems to be safe and effective in improving functional outcome.⁵⁸ Similarly, the STRATIS registry of 1000 patients with anterior-circulation LVO between 2014 and 2016 found a trend toward higher rates of substantial reperfusion and good functional outcome, and lower mortality in patients receiving rescue IA alteplase.⁵⁹ There are scarce data on the use of tenecteplase in rescue IAT. The pilot ALLY prospective nonrandomized study recruited 20 patients to receive up to 4.5 mg of IA tenecteplase and observed a favorable clinical outcome in 50% of patients while 1 patient developed sICH.⁶⁰

FUTURE DIRECTIONS

Despite uncertainties and mixed results, recent studies have provided considerable evidence supporting the efficacy and safety of adjunctive IAT after successful MT in a selected cohort of patients. The most recent American Heart Association guidelines included IAT after successful MT as a potentially reasonable treatment before the positive ANGEL-TNK and PEARL trials were published for incorporation into the guideline review.⁶¹ The

subsequent report of the CHOICE 2 trial conducted in a European population provides additional support for this guideline recommendation.

Given the heterogeneity in patient selection between studies, further research identifying the most appropriate candidates for adjunctive IAT is vital. Pooled analyses at the individual participant level with harmonized end point definitions and core laboratory imaging analysis, particularly in the assessment of tissue reperfusion, will be essential to explore subgroup interactions and secondary end points that likely require a larger sample size to detect small, minimally clinically important differences.³⁴

Three recently completed RCTs will provide insight into the broader applications of IAT. The INSIST-IT trial (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05657457), which tested the benefit of high-dose IAT tenecteplase (0.25 mg/kg, maximum 25 mg) in acute basilar occlusion with successful MT (mTICI 2b–3), was terminated due to potential harm. The BRETIS-TNK 2 trial (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05657444) investigated the use of peri-retrieval high-dose tenecteplase infusion (up to 12 mg) administered before, during, and after first-pass retrieval. The RESCUE-TNK trial (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05657470) tested the effect of IAT tenecteplase in primary medium vessel occlusion or residual medium vessel occlusion after MT for LVO.

Among ongoing studies for anterior-circulation LVO (Table S2), the POST-UK2 (ChiCTR2500105159) trial adopts a pragmatic design and tests IA UK in a broadened angiographic inclusion criterion of eTICI 2b to 3 for non-large-artery atherosclerotic stroke. The ALLY2 (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06949228) will test IA tenecteplase in a North American population. The PEARL-TNK (ChiCTR2600117118) will evaluate adjunctive IA tenecteplase in patients with eTICI 2b reperfusion at an individualized dosage based on the use of bridging IVT (0.125 mg/kg tenecteplase if IVT was not given and 0.0625 mg/kg TNK with IVT). The phase 3 EXTEND-AGNES TNK (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05892510) and IA-SUCCESS (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06768138) will test the use of IAT tenecteplase (0.0625 mg/kg) and alteplase (0.225 mg/kg), including patients with IVT and those presenting with large ischemic core (ASPECTS ≥ 3) in anterior-circulation LVO. The EXTEND-AGNES TNK trial adopts a double-blinded, placebo-controlled design, is powered for a primary outcome of mRS score of 0 to 2, and will also test the use of rescue IAT in unsuccessful MT. Finally, PEARL-MeVO (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07137832) will evaluate IA tenecteplase for primary medium vessel occlusion.

IAT was administered empirically in recent trials even when DSA appeared normal. An alternative approach involves selecting patients with demonstrable imaging biomarkers for patient selection. A meta-analysis of recent trials suggests the benefit of IAT is greater in patients with lower grades of reperfusion. However, qualitative eTICI assessment on 2-dimensional DSA lacks sensitivity in detecting postthrombectomy residual perfusion deficits compared with conventional CT and magnetic resonance imaging perfusion. Perfusion imaging on newer-generation flat-panel detector CT hardware performed immediately postthrombectomy inside the angiography suite may complement tissue perfusion evaluation in real time to enable periprocedural decision-making. The phase 2 TNK-FLOW (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06781385) will test IAT tenecteplase in 40 patients with postthrombectomy perfusion deficit identified on flat-panel CTP. In addition, the PROPHET prospective imaging study (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06691230) will evaluate the additional benefit of flat-panel CTP implemented in acute patient workflow as compared with standard DSA. However, flat-panel CT perfusion technology requires an additional contrast-enhanced acquisition, hardware equipment, and postprocessing software packages which are not widely available in the majority of thrombectomy centers. Emerging techniques to generate tissue perfusion imaging biomarkers from DSA acquisition at the time of thrombectomy may provide a cost-effective option to instantaneously identify persistent perfusion deficit without additional equipment, procedural time, radiation exposure, or contrast administration.^{62,63} (Figure S2).

Given the complex multifactorial pathophysiologic mechanisms underlying postthrombectomy hypoperfusion, fibrinolysis of existing thrombi alone may be insufficient. The positive results of the ATTRACTION trial, reporting an increased likelihood of functional independence with postthrombectomy combined IA and intravenous tirofiban, illustrate the potential importance of mitigating subsequent platelet-driven microvascular thrombosis or re-occlusion.⁶⁴ Future research may explore the development of adjunctive multitargeted agents or combination treatments to restore microvascular reperfusion following mechanical thrombectomy.

CONCLUSIONS

The role of IAT in ischemic stroke has evolved substantially in the past 3 decades—from a candidate primary endovascular treatment, to an off-label rescue after failed endovascular intervention, to an adjunctive therapy to augment the treatment benefit of MT. In the context of an improved understanding of postrecanalization pathophysiology, recent trials provide compelling

reasons to cautiously contemplate IAT as an option in the armamentarium for acute LVO. Nonetheless, the recognized heterogeneity and limitations in study design, and observed mixed results call for additional high-quality, large-scale studies supported by a well-designed, coordinated effort to pool individual patient data for a definitive assessment of the role of IAT. Ongoing clinical trials will provide important insights into unresolved questions regarding optimal patient selection to inform regulatory approval and implementation into clinical practice. The next advance in reperfusion treatment after intravenous thrombolysis and MT may be imminent.

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Supplemental Material

Tables S1–S2

Figures S1–S2



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